

AI-Based Student Performance and Placement Prediction

Problem Statement

Many students are uncertain about their placement readiness because academic performance alone does not determine placement success. Factors such as attendance, study hours, internships, certifications, aptitude, communication skills, coding skills, and project experience all influence employability.

This project aims to develop an AI-powered machine learning solution using IBM watsonx.ai AutoAI on IBM Cloud that predicts whether a student is likely to be placed based on academic and skill-related parameters.

The model is trained on a structured dataset containing features such as Attendance, Study Hours, CGPA, Projects, Internship Experience, Certifications, Aptitude Score, Communication Skills, and Coding Skills. IBM AutoAI automatically prepares the data, performs feature engineering, trains multiple machine learning pipelines, selects the best-performing model, and deploys it as an online API.

The solution helps students understand their placement readiness, identify areas that need improvement, and enables educational institutions to provide timely career guidance through data-driven insights.

Expected Outcomes

- Predict placement likelihood with high accuracy.
- Help students improve employability.
- Assist colleges in placement planning.
- Demonstrate end-to-end AI model development and deployment using IBM Cloud.